

FPA Assistant — System Specification & User Manual

Official AI-Powered Student Information & Administrative Helpdesk Service

System Name: FPA Assistant (v1.5.0)

Author / Lead: AJALA JOSHUA OLUWAFERANMI (FPA/CS/24/3-0089)

Admin Email: joshua@ajala.com

Production Status: ● Fully Active & Verified

Knowledge Base: 55 Verified Records across 11 Categories

Primary AI Model: Google Gemini (Sub-Second Multi-Model Cascade)

1. EXECUTIVE SUMMARY & CORE VALUE

FPA Assistant is an enterprise-grade conversational AI and administrative helpdesk platform engineered specifically for The Federal Polytechnic, Ado-Ekiti. Sourced directly from official institutional circulars and the school portal (fedpolyado.edu.ng), the system provides students, prospective candidates, and faculty with instantaneous, verified answers to academic and administrative inquiries 24 hours a day, 7 days a week.

The software features an official institutional visual identity (forest green #004D2E), zero cartoon emojis, zero generic marketing buzzwords, sub-second query latency (~0.62 seconds), and complete privacy controls for administrative governance.

2. KNOWLEDGE BASE ADMINISTRATION (CAN THE ADMIN STILL POPULATE IT?)

YES, THE ADMINISTRATOR HAS FULL DYNAMIC CONTROL OVER THE KNOWLEDGE BASE:

The administrator can continuously populate, modify, or retire institutional knowledge records at any time through the built-in web portal without touching any code or restarting the server.

How the Administrator Populates & Manages the Knowledge Base:

1. **Sign In as Administrator:** Navigate to `http://localhost:5000/login` and enter `joshua@ajala.com` (Password: Admin123!).
2. **Access the Admin Panel:** Click **Admin Panel** in the top header (or visit `http://localhost:5000/admin`).
3. **Create New Entry:** Click the "+ Add Knowledge Entry" button. A modal window opens requesting:
 - **Service Category:** Choose from 11 categories (Admission, Fees, Registration, Exams, Hostel, etc.).
 - **Question / Topic:** The student inquiry (e.g., *"What is the deadline for 2026/2027 acceptance fee payment?"*).
 - **Official Answer:** Detailed verified institutional response (supports Markdown bolding, lists, and links).
 - **Keywords:** Comma-separated search tags (e.g., acceptance fee, deadline, 2026).
 - **Source Reference:** The issuing authority (e.g., *"Admissions Directorate Official Bulletin"*).
 - **Status:** *Active* (immediately available to the AI) or *Inactive*.
4. **Instant Live Availability:** Once saved, the entry is immediately stored in PostgreSQL and is instantly queried by the AI engine. No re-indexing or server reboot is needed.
5. **Editing & Deletion:** Admins can update figures, deadlines, or portal URLs anytime by clicking **Edit** on any record row, or remove obsolete records by clicking **Delete**.
6. **Bulk SQL Ingestion:** For batch updates, records can also be appended to `database/schema/knowledge_base_expanded_seed.sql`.

3. COMPREHENSIVE FUNCTIONALITIES OF FPA ASSISTANT

A. Student Helpdesk & Conversational AI Experience

- **Sub-Second Conversational Response:** Delivers intelligent, natural responses in ~0.62 seconds using Google Gemini (gemini-flash-lite-latest and gemini-3.7-flash).
- **Natural Greeting & Intent Interception:** Automatically detects casual greetings (*"hello"*, *"good morning"*), identity questions (*"who are you"*), and gratitude (*"thank you"*) and responds warmly without dumping raw FAQs.
- **Strict, Question-Focused Answering:** The AI responds *only* to the exact question asked by the student,

completely eliminating boilerplate data dumps or unrequested FAQs.

- **Multi-Turn Context & Follow-up Memory:** Tracks recent conversation turns so students can ask follow-ups naturally (e.g., *"How much is it?"* after asking about acceptance fees).
- **Rich Document Formatting:** Outputs formatted bold headers, bullet lists, numbered procedural steps, and clickable links to official portal pages (<https://students.fedpolyado.edu.ng> and <https://fedpolyado.edu.ng>).
- **Interactive Student Feedback Loop:** Students can rate answers with one click (*"Yes, accurate"* or *"Incomplete"*), saving records to the feedback analytics table.
- **Quick Inquiry Chips:** Common campus questions are displayed as interactive buttons on empty chat sessions for immediate one-click answers.
- **Guest vs. Registered Mode:** Guests can ask questions freely without registration; logged-in students have their sessions automatically preserved.

B. Student Accounts & Session History

- **Student Self-Registration:** Students can create personal accounts directly via `/login` with name, email, and password.
- **Encrypted Security:** Passwords are encrypted using bcrypt (10 rounds); sessions are secured using signed 7-day JSON Web Tokens (JWT).
- **Persistent Session Drawer:** Authenticated students can toggle the **"Session History"** sidebar to review past discussions, switch between topics, or start a new inquiry.

C. Campus Services Overview & Directory

- **Campus Directory Grid:** Landing page highlights 6 primary operational areas (Admissions, School Fees & Remita, Course Registration, Examinations & MIS Results, Hostel Accommodation, and ICT Support).
- **One-Click Pre-Filled Queries:** Clicking any service card launches the helpdesk with the verified inquiry pre-filled.
- **Official Institutional Identity:** Professional non-AI-slop design system built with official Federal Polytechnic forest green (#004D2E), clean typography, and SVG vector icons.

D. Administrative Governance & Control Panel (`/admin`)

- **Hidden & Restricted Access:** The Admin Portal and Knowledge Base Explorer are completely hidden from regular users and students; access is restricted exclusively to `joshua@ajala.com`.
- **Live Institutional Analytics:** Real-time KPI cards displaying Total Registered Students, Total Inquiries Processed, Total Sessions, and Total Active Knowledge Records.
- **Full Knowledge Base CRUD:** Create, Read, Update, and Delete institutional records with category and status filters.
- **Knowledge Base Explorer (`/knowledge-base`):** Private, full-text searchable catalog across all 55 verified entries for administrative inspection.

E. AI Microservice & NLP Architecture

- **Intent Classification:** Categorizes incoming student queries into 11 institutional categories using spaCy NLP.
- **Multi-Model Gemini Cascade:** Automatically tries high-speed Gemini models with 10-second timeout guards.
- **Resilient Deterministic Fallback:** If cloud networks are temporarily interrupted, the system automatically scores the 55 database records to extract the best matching answer directly, ensuring zero downtime.

F. Production Operations & Orchestration

- **Unified Single-Port Delivery:** Node.js Express on port 5000 delivers both the REST API and the compiled React production bundle (frontend/dist).
- **Lifecycle Scripts:** ./start-services.sh, ./status-services.sh, and ./stop-services.sh provide 1-command service management.
- **Automated Testing Suite:** Built-in 14-point test suite (node backend/src/e2e_verification.js) covering health, authentication, CRUD, and chat persistence.
- **Cloud Shipping Ready:** Complete instructions for deployment on Cloud PaaS (Render/Railway) or an Ubuntu Cloud VPS with Nginx and SSL.

4. INSTITUTIONAL KNOWLEDGE BASE TAXONOMY (55 VERIFIED RECORDS)

ID	Category Name	Records	Verified Institutional Scope & Topics Covered
1	Admission	10	ND Full-Time cut-offs, Post-UTME screening, Part-Time/Evening entry, HND entry (Lower Credit + 1-yr IT, Pass + 2-yr IT), non-refundable ₦45,000 acceptance fee, overview of 6 academic schools.
2	Course Registration	6	Semester portal registration procedures, credit unit boundaries (15 min – 24 max), Add/Drop window, Course Adviser and HOD approvals.
3	School Fees	7	Remita invoice generation, RRR payment verification on portal, resolving unconfirmed bank debits, semester installments, Bursary & ICT emails.
4	Examination	6	75% lecture attendance requirement for exam eligibility, carryover retakes, NBTE standard grading scale (Distinction, Upper Credit, Lower Credit, Pass, Fail).
5	Academic Calendar	5	15-week semester structure, Rector Engr. Dr. Temitope John Alake , Registry, Bursary, Principal Officers, matriculation, and convocation.
6	Hostel Services	3	Abuja Hall of Residence , Student Affairs allocation, safety regulations,

ID	Category Name	Records	Verified Institutional Scope & Topics Covered
			prohibited electrical appliances (boiling rings, hot plates).
7	SIWES	3	4-month mandatory practical attachment for ND I transitioning to ND II, ITF logbook maintenance, grading and departmental defense.
8	Library Services	3	Central Polytechnic Library location, opening hours (Mon–Fri 8AM–6PM, Sat 9AM–2PM), borrower registration, loan duration.
9	ICT Support	6	Student portal telephone support (07088391544, 09083892022), support emails (support@lloydant.com, ict@fedpolyado.edu.ng), password recovery.
10	Transcript Services	2	Online transcript application via students.fedpolyado.edu.ng, RRR courier fee payment, domestic and international institutional dispatch.
11	Graduation Requirements	4	Minimum 2.00 CGPA, institutional final clearance across 6 units (Department, School, Library, Student Affairs, Bursary, Security/Alumni), NYSC mobilization.

5. AUTOMATED QUALITY ASSURANCE & VERIFICATION AUDIT

The system was audited via the automated 14-point test suite (node backend/src/e2e_verification.js):

Test Suite Item	Target Endpoint	Audit Result
Backend Service Availability	GET /health	PASS (200 OK)
AI Microservice Operational Health	GET :5001/health	PASS (200 OK)
Category Directory Retrieval (11 Categories)	GET /api/categories	PASS (11 Rows)
Knowledge Base Search & Filtering	GET /api/kb	PASS (200 OK)
Administrator Authentication (joshua@ajala.com)	POST /api/auth/login	PASS (Admin JWT)
Live Institutional Analytics Retrieval	GET /api/admin/analytics	PASS (KPIs verified)
Admin Create Knowledge Base Record	POST /api/admin/kb	PASS (Record created)
Admin Update Knowledge Base Record	PUT /api/admin/kb/:id	PASS (Record updated)
Admin Delete Knowledge Base Record	DELETE /api/admin/kb/:id	PASS (Record deleted)
Student Self-Registration	POST /api/auth/register	PASS (User created)
Multi-Turn Chat Persistence & AI Response	POST /api/chat/message	PASS (Message ID assigned)
Conversation History Retrieval	GET /api/conversations	PASS (History restored)

Test Suite Item	Target Endpoint	Audit Result
Student Response Feedback Submission	POST /api/chat/feedback	PASS (Rating stored)
Unified Single Page Application Serving	GET /	PASS (SPA Delivered)

Document Ref: FPA-ASST-SPEC-2026 The Federal Polytechnic, Ado-Ekiti (Ekiti State, Nigeria) Certified Production Ready